

Vishal Kumar Allam

Dallas, TX | 816-666-2813 | itsmevishal36@gmail.com | **Work Authorization: H1B**

PROFESSIONAL SUMMARY

- Results-driven **Forward Deployed Engineer / Software Engineer** with **6+ years of experience** designing, implementing, integrating, and supporting distributed data platforms, AI/ML infrastructure, hybrid-cloud environments, and mission-critical enterprise applications.
- Strong expertise architecting **AI/ML and HPC data environments** that provide scalable, low-latency access to large datasets across distributed storage, GPU compute clusters, on-premises infrastructure, and public cloud platforms.
- Hands-on experience with **GPU orchestration, NVIDIA and Mellanox technologies, InfiniBand, RoCE, 100GbE+ networking, NFS, SMB, S3, distributed file systems, and high-performance storage architectures.**
- Production coding and scripting expertise using **Python, Go, C++, REST APIs, Linux shell scripting, Kubernetes, and Docker** to build custom integrations, automation tools, operational utilities, and last-mile solutions for complex customer environments.
- Deep Linux and systems knowledge including **kernel tuning, storage stack optimization, network troubleshooting, routing, latency analysis, throughput optimization, containerization, and distributed-system performance debugging.**
- Experienced operating as a customer-facing technical authority, translating business and infrastructure challenges into scalable solution architectures while collaborating with data scientists, IT administrators, engineering teams, executives, and customer success organizations.
- Autonomous problem solver with experience identifying customer friction points, converting field-discovered issues into reusable platform improvements, recognizing expansion opportunities, and driving successful deployments in ambiguous and high-stakes environments.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Automation:** Python, Go, C++, Bash/Shell Scripting, REST APIs, JSON, Automation Scripts, Production-Grade Integration Tools
- **AI/HPC Infrastructure:** AI/ML Training Pipelines, HPC Workloads, GPU Orchestration, NVIDIA GPUs, Mellanox Networking, Distributed Training, Low-Latency Data Access, High-Throughput Computing
- **High-Speed Networking:** InfiniBand, RoCE, RDMA, 100GbE+, TCP/IP, Routing, Network Latency Analysis, Throughput Optimization, Distributed Network Troubleshooting
- **Storage & Data Platforms:** NFS, SMB, S3, Distributed File Systems, Unified Namespace, Global Data Environment, Hybrid-Cloud Storage, Data Mobility, Data Orchestration, Hammerspace Architecture
- **Linux & Containers:** Linux Internals, Kernel Tuning, Storage Stack Performance, Filesystems, Process/Memory Analysis, Kubernetes, Docker, Containerized Workloads
- **Cloud & Integration:** AWS, Azure, Google Cloud, Hybrid Cloud, Enterprise APIs, Storage APIs, Cloud Object Storage, Custom Integrations, Infrastructure Automation
- **Forward-Deployed Engineering:** Solution Architecture, Deep Customer Embedding, Production Troubleshooting, Technical Discovery, Stakeholder Management, Business Opportunity Identification, Engineering Feedback Loops, Agile Delivery

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Acted as a forward-deployed technical engineer for strategic enterprise environments, working directly with customer-facing teams to identify infrastructure friction points, design solutions, and drive complex implementations from discovery through production stabilization.
- Designed hybrid-cloud data architectures supporting **AI/ML and high-performance computing workloads**, providing scalable and low-latency access to large datasets across distributed storage, cloud object stores, and accelerated compute environments.
- Engineered data-access and storage integrations using **NFS, SMB, S3, REST APIs, distributed filesystems, and unified namespace patterns**, enabling applications and compute workloads to access geographically distributed datasets through consistent interfaces.
- Developed production-grade automation tools and custom integrations using **Python, Go, C++, Bash, and REST APIs** to bridge product capabilities with customer-specific compute, storage, networking, and operational infrastructure.
- Optimized GPU-intensive workloads across **NVIDIA and Mellanox environments**, troubleshooting InfiniBand, RoCE, RDMA, and 100GbE+ networking issues involving routing, packet flow, latency, congestion, and throughput bottlenecks.
- Performed deep Linux systems troubleshooting and performance optimization involving kernel parameters, filesystem behavior, storage I/O paths, memory/process analysis, network stacks, Kubernetes clusters, and Docker-based application environments.
- Partnered with sustaining and core engineering teams to reproduce field-discovered issues, perform root-cause analysis, develop interim solutions, and translate recurring customer problems into reusable product enhancements and platform capabilities.
- Communicated technical findings and architectural recommendations to data scientists, infrastructure engineers, IT administrators, Product teams, Customer Success, and executive stakeholders while identifying adjacent workloads and expansion opportunities within strategic accounts.

Environment: Python, Go, C++, Bash, Linux, NVIDIA, Mellanox, GPU Orchestration, AI/ML Pipelines, HPC, InfiniBand, RoCE, RDMA, 100GbE+, NFS, SMB, S3, Distributed File Systems, Unified Namespace, REST APIs, Kubernetes, Docker, AWS, Azure, Hybrid Cloud, Performance Tuning, Network Troubleshooting, Solution Architecture

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and supported distributed data and compute infrastructure for large-scale healthcare and AI workloads, working closely with engineering and platform teams to resolve performance, integration, and data-access challenges across hybrid environments.
- Architected high-throughput data paths supporting **AI/ML training and analytics pipelines**, optimizing movement of large healthcare datasets between object storage, distributed filesystems, containerized applications, and accelerated GPU compute resources.
- Integrated enterprise storage and application environments using **NFS, SMB, S3, REST APIs, and distributed data-access patterns**, providing consistent access to datasets across on-premises and cloud infrastructure.
- Developed Python, Go, C++, and shell-based utilities for data movement, infrastructure automation, API integrations, workload orchestration, health checks, monitoring, and operational troubleshooting across production environments.
- Supported **NVIDIA GPU and Mellanox networking infrastructure**, analyzing InfiniBand, RoCE, RDMA, 100GbE+ routing, latency, congestion, and throughput characteristics affecting distributed AI and high-performance data workloads.
- Troubleshot Linux and container infrastructure at system level, including kernel configuration, filesystem performance, network interfaces, memory and CPU utilization, storage stack behavior, Kubernetes scheduling, and Docker runtime issues.
- Collaborated with Software Engineering, Infrastructure, DevOps, Data Science, Product, and Customer-facing teams to translate workload requirements into practical architectures and convert production findings into reusable automation and engineering improvements.

- Presented complex infrastructure constraints, root causes, performance findings, and remediation options to technical and non-technical stakeholders while proactively identifying opportunities to expand platform usage across additional data and AI workloads.

Environment: Python, Go, C++, Linux, NVIDIA GPUs, Mellanox, InfiniBand, RoCE, RDMA, 100GbE+, AI/ML Data Pipelines, HPC, NFS, SMB, S3, Distributed Storage, REST APIs, Kubernetes, Docker, AWS, Azure, Hybrid Cloud, Network Optimization, Storage Performance, Infrastructure Automation

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Developed backend software, automation utilities, and enterprise integrations using Python, Java, C++, SQL, REST APIs, and Linux-based development environments for distributed business applications.
- Designed data-access and integration workflows connecting applications with enterprise storage systems, APIs, relational databases, filesystems, and cloud-hosted services while maintaining performance and reliability.
- Worked extensively with Linux systems to troubleshoot process, memory, filesystem, network, and application-level issues while tuning configurations for improved application and storage performance.
- Developed reusable automation scripts for environment configuration, data movement, service validation, deployment, monitoring, and operational troubleshooting across development and production environments.
- Supported containerized workloads using Docker and Kubernetes concepts, validating application behavior, service connectivity, resource utilization, persistent storage, and distributed application dependencies.
- Investigated latency and throughput issues across APIs, databases, network services, and application components using logs, metrics, system-level diagnostics, and performance-analysis techniques.
- Collaborated with architects, Product Managers, Software Engineers, DevOps teams, and business stakeholders to translate technical requirements into maintainable integrations and production-ready software solutions.
- Provided production support and root-cause analysis for distributed application failures, documented operational procedures, and contributed reusable fixes and engineering improvements to future releases.

Environment: Python, Java, C++, SQL, Linux, REST APIs, Distributed Systems, Enterprise Storage, Docker, Kubernetes, Networking, Performance Tuning, Shell Scripting, Automation, AWS, Git, Jenkins, CI/CD, Agile Scrum

Associate Software Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and supported enterprise software components, integrations, and automation utilities using Java, Python, SQL, REST APIs, shell scripting, and Linux-based application environments.
- Integrated frontend and backend applications with databases, filesystems, APIs, and distributed services while troubleshooting connectivity, data-flow, and application-performance issues across multi-tier architectures.
- Created reusable scripts and engineering utilities supporting deployment validation, data processing, environment configuration, service monitoring, and production troubleshooting activities.
- Worked with Linux systems and networking fundamentals to analyze process behavior, filesystem issues, application logs, service connectivity, resource utilization, latency, and throughput problems.
- Supported containerized and CI/CD-driven application deployments using Docker, Jenkins, Maven, Git, and automated validation workflows across development and production environments.
- Performed integration and system-level validation across REST APIs, databases, backend services, and distributed components to identify complex defects and isolate infrastructure-related root causes.
- Collaborated with Software Engineers, infrastructure teams, business analysts, QA leads, and Product stakeholders to resolve production issues and translate customer and operational problems into actionable engineering improvements.
- Prepared technical documentation, troubleshooting procedures, implementation notes, integration specifications, and production-support guidance while contributing to Agile delivery and continuous improvement initiatives.

Environment: Java, Python, SQL, Linux, Bash, REST APIs, Distributed Systems, Networking, Filesystems, Docker, Jenkins, Maven, Git, CI/CD, Production Support, Performance Troubleshooting, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.